

Gun_Violence_Against_Women_NY(2006-2024)

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2026-09-22

Aim of the report

Using the dataset titled *NYPD Shooting Incident Data (Historic)* available at the <https://catalog.data.gov/dataset>, this report imports, cleans, and analyzes the data to investigate gun violence against women.

Specifically, the analysis aims to answer the following question

- Do the shootings involving women occur in different locations (e.g. inside dwellings vs. open streets) or at different hours compared to male victims?

Each record in the dataset represents a shooting incident in New York City from 2006 to 2024. The data includes details about the event, its location and the time of occurrence, alongside demographic information for both suspects and victims.

Required Packages

Please install the below list of packages that are used in this analysis.

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.2.1      v readr      2.2.0
## v forcats    1.0.1      v stringr   1.6.0
## v ggplot2    4.0.3      v tibble    3.3.1
## v lubridate  1.9.5      v tidyr     1.3.2
## v purrr      1.2.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

Import the Data

Import the data from the source url.

```
url <- 'https://data.cityofnewyork.us/api/v3/views/833y-fsy8/export.csv?accessType=DOWNLOAD'
nypd_shooting_data <- read.csv(url)
```

View the Data

Look at how the data is currently to know what needs to be done to clean and transform it for visualization and analysis.

```
dplyr::glimpse(nypd_shooting_data)
```

```
## Rows: 29,744
## Columns: 21
## $ INCIDENT_KEY      <int> 298699604, 298699604, 298672096, 298672094, 29~
## $ OCCUR_DATE        <chr> "12/31/2024", "12/31/2024", "12/30/2024", "12/~
## $ OCCUR_TIME        <chr> "19:16:00", "19:16:00", "16:45:00", "12:15:00"~
## $ BORO              <chr> "BROOKLYN", "BROOKLYN", "BRONX", "BRONX", "BRO~
## $ LOC_OF_OCCUR_DESC <chr> "OUTSIDE", "OUTSIDE", "OUTSIDE", "OUTSIDE", "I~
## $ PRECINCT          <int> 69, 69, 47, 52, 41, 47, 60, 47, 23, 43, 73, 18~
## $ JURISDICTION_CODE <int> 0, 0, 0, 0, 0, 0, 2, 0, 0, 0, 0, 0, 0, 0, 0~
## $ LOC_CLASSFCTN_DESC <chr> "STREET", "STREET", "STREET", "STREET", "DWELL~
## $ LOCATION_DESC     <chr> "(null)", "(null)", "(null)", "(null)", "MULTI~
## $ STATISTICAL_MURDER_FLAG <chr> "false", "false", "false", "false", "true", "f~
## $ PERP_AGE_GROUP    <chr> "25-44", "25-44", "(null)", "45-64", "18-24", ~
## $ PERP_SEX          <chr> "M", "M", "(null)", "M", "M", "(null)", "M", "~
## $ PERP_RACE         <chr> "BLACK", "BLACK", "(null)", "BLACK", "BLACK", ~
## $ VIC_AGE_GROUP     <chr> "25-44", "18-24", "<18", "25-44", "25-44", "25~
## $ VIC_SEX           <chr> "M", "M", "F", "M", "M", "F", "M", "M", "M", "~
## $ VIC_RACE          <chr> "BLACK", "BLACK", "WHITE HISPANIC", "WHITE", "~
## $ X_COORD_CD        <chr> "1,015,120", "1,015,120", "1,021,316", "1,017,~
## $ Y_COORD_CD        <chr> "173,870", "173,870", "259,277", "260,875", "2~
## $ Latitude          <dbl> 40.64387, 40.64387, 40.87826, 40.88266, 40.827~
## $ Longitude         <dbl> -73.88876, -73.88876, -73.86596, -73.87896, -7~
## $ Lon_Lat           <chr> "POINT (-73.888761 40.643866)", "POINT (-73.88~
```

```
summary(nypd_shooting_data)
```

```
## INCIDENT_KEY      OCCUR_DATE      OCCUR_TIME      BORO
## Min.   : 9953245   Length :29744   Length :29744   Length :29744
## 1st Qu.: 67321140  N.unique : 6431  N.unique : 1620  N.unique : 5
## Median :109291972  N.blank  : 0    N.blank  : 0    N.blank  : 0
## Mean   :133850951  Min.nchar: 10   Min.nchar: 7    Min.nchar: 5
## 3rd Qu.:214741917  Max.nchar: 10   Max.nchar: 8    Max.nchar: 13
## Max.   :299462478
##
## LOC_OF_OCCUR_DESC  PRECINCT      JURISDICTION_CODE LOC_CLASSFCTN_DESC
## Length :29744     Min.   : 1.00   Min.   :0.0000   Length :29744
## N.unique : 3      1st Qu.: 44.00 1st Qu.:0.0000   N.unique : 11
## N.blank :25596     Median : 67.00 Median :0.0000   N.blank :25596
## Min.nchar: 0      Mean   : 65.23 Mean   :0.3181   Min.nchar: 0
## Max.nchar: 7      3rd Qu.: 81.00 3rd Qu.:0.0000   Max.nchar: 11
## Max.   :123.00    Max.   :2.0000
## NAs :2
## LOCATION_DESC     STATISTICAL_MURDER_FLAG PERP_AGE_GROUP    PERP_SEX
## Length :29744     Length :29744   Length :29744   Length :29744
## N.unique : 41      N.unique : 2     N.unique : 13    N.unique : 5
## N.blank :14977     N.blank  : 0     N.blank  : 9344  N.blank  : 9310
```

```
## Min.nchar:    0   Min.nchar:    4           Min.nchar:    0   Min.nchar:    0
## Max.nchar:   25   Max.nchar:    5           Max.nchar:    7   Max.nchar:    6
##
##
##      PERP_RACE      VIC_AGE_GROUP      VIC_SEX      VIC_RACE
## Length :29744   Length :29744   Length :29744   Length :29744
## N.unique :    9   N.unique :    7   N.unique :    3   N.unique :    7
## N.blank  : 9310   N.blank  :    0   N.blank  :    0   N.blank  :    0
## Min.nchar:    0   Min.nchar:    3   Min.nchar:    1   Min.nchar:    5
## Max.nchar:   30   Max.nchar:    7   Max.nchar:    1   Max.nchar:   30
##
##
##      X_COORD_CD      Y_COORD_CD      Latitude      Longitude
## Length :29744   Length :29744   Min. :40.51   Min. : -74.25
## N.unique :13897   N.unique :14043   1st Qu.:40.67   1st Qu.: -73.94
## N.blank  :    0   N.blank  :    0   Median :40.70   Median : -73.91
## Min.nchar:    6   Min.nchar:    6   Mean :40.74   Mean : -73.91
## Max.nchar:   23   Max.nchar:   22   3rd Qu.:40.83   3rd Qu.: -73.88
##                                     Max. :40.91   Max. : -73.70
##                                     NAs :97      NAs :97
##
##      Lon_Lat
## Length :29744
## N.unique :14058
## N.blank  :   97
## Min.nchar:    0
## Max.nchar:   45
##
##
```

Tidy the Data

With the included tidyverse package functions, clean and transform the data. From the imported `nypd_shooting_data` dataframe

1. Remove the unwanted columns for our analysis like `X_COORD_CD`, `Y_COORD_CD`, `Latitude`, `Longitude`, `Lon_lat`
2. Convert the `OCCUR_DATE` and `OCCUR_TIME` columns from char data type to date and period to query based on date and time
3. Rename the `LOC_CLASSFCTN_DESC` to `LOCATION` as that is the location related column we will be primarily using for the analysis
4. Convert both the empty strings and “(null)” text values in the `LOCATION` to NA values
5. Lets not consider the rows with missing data for `LOCATION` for our analysis. So lets filter them out.
6. Similarly lets filter out the rows with `VIC_SEX` as “U” (unknown)
7. For our analysis on time of violence, lets create a new data frame with field `OCCUR_HOUR` to group the `OCCUR_TIME` by the hour of occurrence

Clean the Data

```
# Remove the fields with co-ordinate info and convert the date/time to correct data type
nypd_shooting_data <- nypd_shooting_data |>
# Remove the fields we will not be using in this analysis
```

```
dplyr::select(-c(LOCATION_DESC:PERP_RACE, X_COORD_CD:Lon_Lat)) |>
# Convert the date/time fields from chr datatype to date and period datatypes
dplyr::mutate(
  OCCUR_DATE = lubridate::mdy(OCCUR_DATE), OCCUR_TIME = lubridate::hms(OCCUR_TIME)
) |>
# Rename the LOC_CLASSFCTN_DESC field to LOCATION
dplyr::rename(LOCATION = LOC_CLASSFCTN_DESC )

# There are two null values - "", "(null)" for the renamed field LOCATION.
unique(nypd_shooting_data$LOCATION)
```

```
## [1] "STREET"      "DWELLING"    "HOUSING"     "OTHER"       "TRANSIT"
## [6] "VEHICLE"     "COMMERCIAL"  "(null)"      "PLAYGROUND"  "PARKING LOT"
## [11] ""
```

```
# Convert both the null values to true R NA values for better handling
nypd_shooting_data$LOCATION[nypd_shooting_data$LOCATION=="(null)" |
                           nypd_shooting_data$LOCATION==""] <- NA
unique(nypd_shooting_data$LOCATION)
```

```
## [1] "STREET"      "DWELLING"    "HOUSING"     "OTHER"       "TRANSIT"
## [6] "VEHICLE"     "COMMERCIAL"  NA            "PLAYGROUND"  "PARKING LOT"
```

Transform the Data

```
# Create a new data frame by filtering out the rows with NA values for LOCATION
# and "U" for VIC_SEX. Also add new field for analyzing based on the hour of
# occurrence
nypd_loc_and_time_data <- nypd_shooting_data |>
dplyr::filter(!is.na(LOCATION), VIC_SEX != "U") |>
dplyr::mutate(OCCUR_HOUR = hour(OCCUR_TIME)) |>
dplyr::mutate(OCCUR_YEAR = year(OCCUR_DATE)) |>
dplyr::select(c(VIC_SEX, BORO, LOCATION, OCCUR_HOUR, everything()))

glimpse(nypd_loc_and_time_data)

# Compare the male-to-female ratio of the filtered dataset against the original
# dataset to make sure the demographic distribution is not disturbed with filtering
full_dataset_proportions <- nypd_shooting_data |>
  filter(VIC_SEX %in% c("M", "F")) |>
  count(VIC_SEX) |>
  mutate(
    Percentage = (n / sum(n)) * 100,
    Dataset = "Full Dataset"
  )

filtered_dataset_proportions <- nypd_loc_and_time_data |>
  filter(VIC_SEX %in% c("M", "F")) |>
  count(VIC_SEX) |>
  mutate(
```

```

    Percentage = (n / sum(n)) * 100,
    Dataset = "Filtered Dataset"
  )
proportion_comparison <- bind_rows(
  full_dataset_proportions,
  filtered_dataset_proportions) |>
  select(Dataset, VIC_SEX, Count = n, Percentage)
print(proportion_comparison)

```

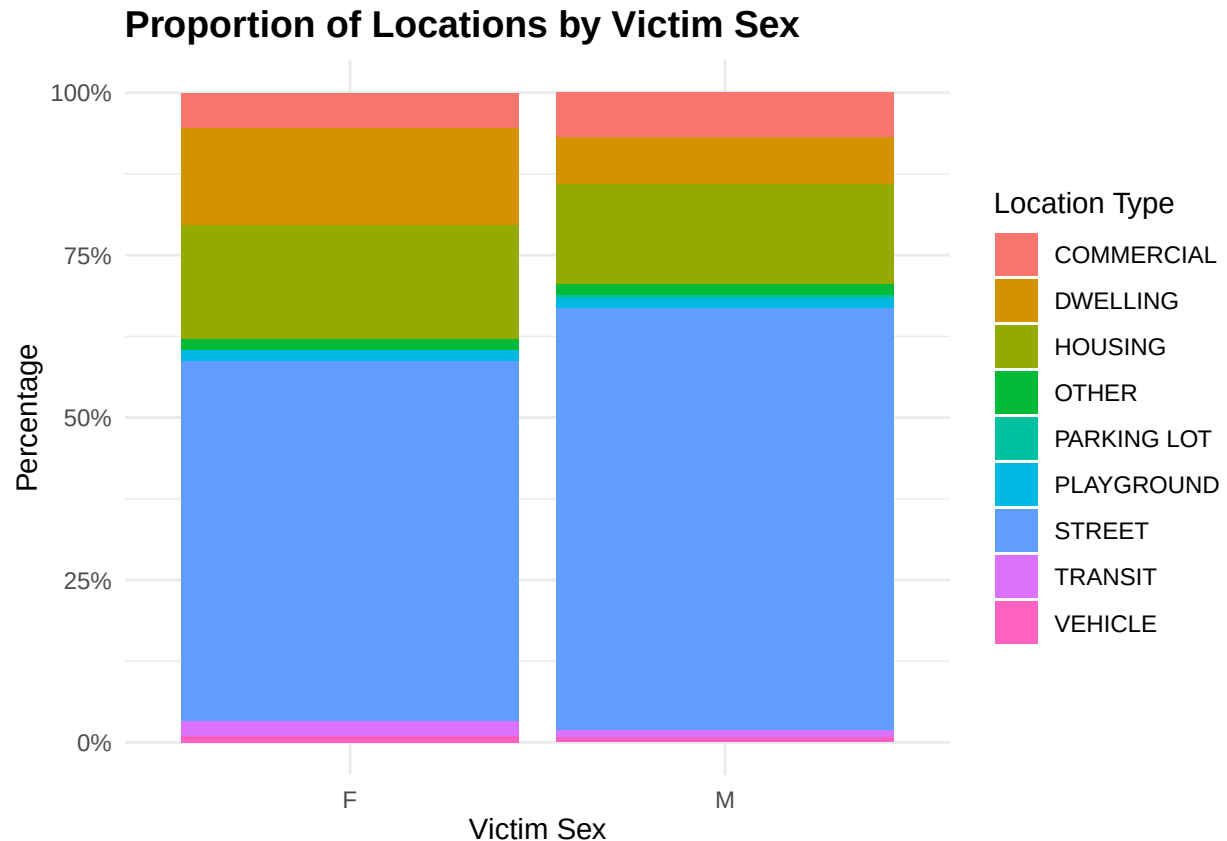
Visualize and Analyze the Data

Now with the transformed data, try to visualize with stacked bar charts to see how the location and hour of incidents compare between female and male victims

```

# Lets plot a Stacked Bar Chart to visualize the Location and Victim Sex data
ggplot2::ggplot(nypd_loc_and_time_data, aes(x = VIC_SEX, fill = LOCATION)) +
  geom_bar(position = "fill") +
  scale_y_continuous(labels = scales::percent) + # Changes y-axis to %
  labs(
    title = "Proportion of Locations by Victim Sex",
    x = "Victim Sex",
    y = "Percentage",
    fill = "Location Type"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(face = "bold", size = 14),
    axis.text.x = element_text(angle = 0, vjust = 0.5)
  )

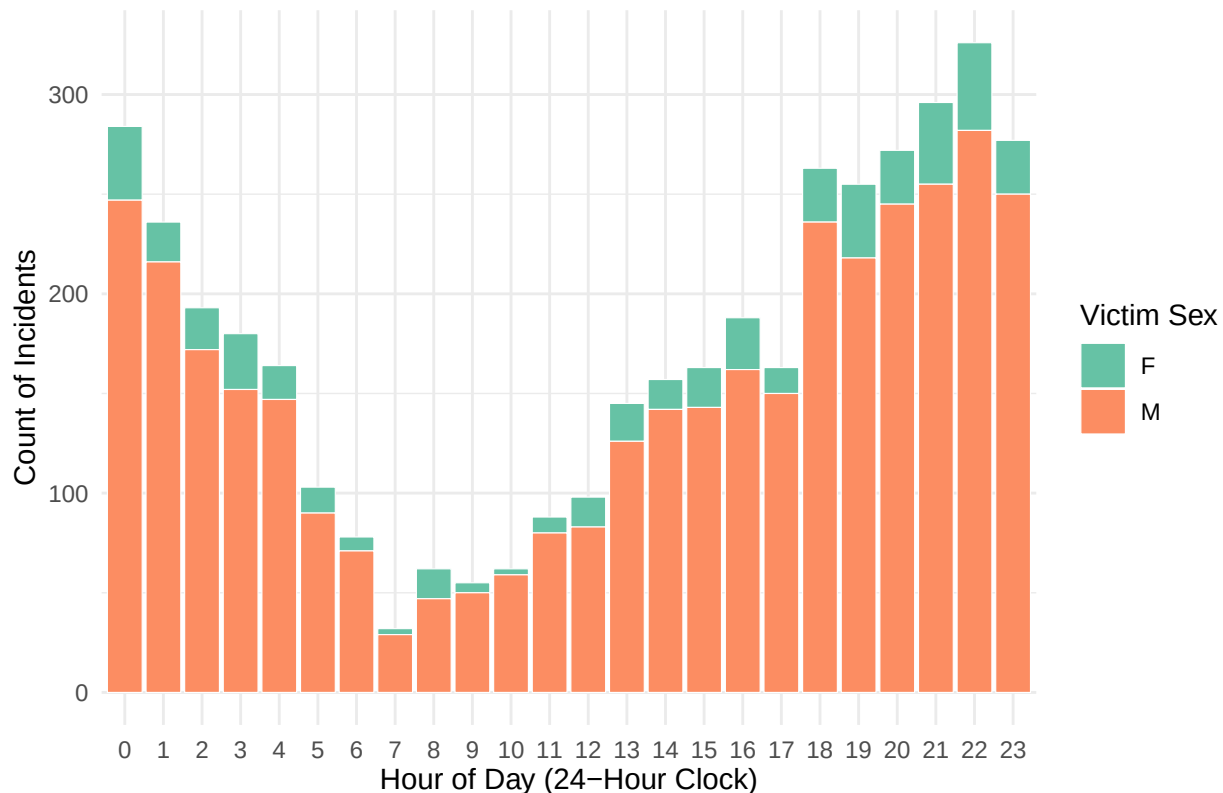
```



From the above chart, we can see that the proportion of gun violence inside the dwellings is higher for female victims compared to male victims. If data were available regarding whether the perpetrator was a known person (such as a family member) or a stranger, we would be able to further analyze this trend. Additionally, the number of incidents in transit is proportionately higher for female victims, suggesting a greater vulnerability in public transportation spaces.

```
# Lets plot a Stacked Bar Chart to visualize the Time and Victim Sex data
ggplot(nypd_loc_and_time_data, aes(x = factor(OCCUR_HOUR), fill = VIC_SEX)) +
  geom_bar(position = "stack", color = "white", linewidth = 0.2) +
  scale_fill_brewer(palette = "Set2") + # Change to professional color palette
labs(
  title = "Incidents by Hour of Occurrence and Victim Sex",
  x = "Hour of Day (24-Hour Clock)",
  y = "Count of Incidents",
  fill = "Victim Sex"
) +
theme_minimal() +
theme(
  plot.title = element_text(face = "bold", size = 14),
  axis.text.x = element_text(angle = 0, vjust = 0.5)
)
```

Incidents by Hour of Occurrence and Victim Sex



From the above chart, we don't see a marked difference in the proportion of incidents between male and female victims with regard to the hour of occurrence. The number of incidents seems to increase after daylight hours in line with the general expectation. Transforming the data by grouping the hour of occurrence into distinct categories (like Late night, Morning, Afternoon, Evening) would make these trends more evident. However, we see a considerable number of incidents against female victims around 8 a.m. and 4 p.m. This could correspond to peak commuting hours. If we were to further analyze the locations of violence during these specific periods, we could determine whether these incidents correspond to those in transit.

Model the Data

Let us see how we can model the data to make predictions about the probability of the victim being a female based on the location, borough and hour of occurrence. Since we are making predictions as to whether the victim is going to be Female or not, we will go with Generalized Linear Model function with family as binomial.

This is going to involve some more transformations to the data.

- First we need to make sure that all the variables/fields used in the formula for prediction is either a numeric vector or factor.
- As our target is female victim, create a new field VIC_SEX_NUM to override R setting of alphabetical first as the baseline otherwise

```
# Convert the fields used in formula to be either numeric vector or factor
model_data <- nypd_loc_and_time_data |>
mutate(
```

```

# 1 if Female, 0 if Male
VIC_SEX_NUM = ifelse(VIC_SEX == "F", 1, 0),
BORO = as.factor(BORO),
LOCATION = as.factor(LOCATION),
# Treat hour as categorical to capture non-linear time of day effects
OCCUR_HOUR = as.factor(OCCUR_HOUR)
)

# Run the model
nypd_model <- glm(
  VIC_SEX_NUM ~ LOCATION + BORO + OCCUR_HOUR,
  data = model_data,
  family = binomial)

summary(nypd_model)

##
## Call:
## glm(formula = VIC_SEX_NUM ~ LOCATION + BORO + OCCUR_HOUR, family = binomial,
##      data = model_data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      -2.22688    0.27916  -7.977  1.5e-15 ***
## LOCATIONDWELLING    0.94784    0.24516   3.866  0.000111 ***
## LOCATIONHOUSING     0.37556    0.23648   1.588  0.112256
## LOCATIONOTHER       0.07046    0.42885   0.164  0.869486
## LOCATIONPARKING LOT -12.48358  217.89856  -0.057  0.954314
## LOCATIONPLAYGROUND   0.40177    0.41884   0.959  0.337430
## LOCATIONSTREET       0.07469    0.21448   0.348  0.727669
## LOCATIONTRANSIT      0.77735    0.41279   1.883  0.059682 .
## LOCATIONVEHICLE      0.53331    0.53115   1.004  0.315350
## BOROBROOKLYN        0.10179    0.12378   0.822  0.410876
## BOROMANHATTAN       0.11014    0.14838   0.742  0.457924
## BOROQUEENS          0.22540    0.15210   1.482  0.138381
## BOROSTATEN ISLAND    0.53727    0.30479   1.763  0.077945 .
## OCCUR_HOUR1         -0.50837    0.29546  -1.721  0.085315 .
## OCCUR_HOUR2         -0.18423    0.29347  -0.628  0.530164
## OCCUR_HOUR3          0.25953    0.27330   0.950  0.342307
## OCCUR_HOUR4         -0.19594    0.31431  -0.623  0.533025
## OCCUR_HOUR5         -0.10478    0.35125  -0.298  0.765459
## OCCUR_HOUR6         -0.40009    0.43839  -0.913  0.361437
## OCCUR_HOUR7         -0.34093    0.63640  -0.536  0.592158
## OCCUR_HOUR8          0.70729    0.35635   1.985  0.047164 *
## OCCUR_HOUR9         -0.35483    0.50663  -0.700  0.483698
## OCCUR_HOUR10        -1.22373    0.62197  -1.967  0.049126 *
## OCCUR_HOUR11        -0.39943    0.41382  -0.965  0.334437
## OCCUR_HOUR12         0.18739    0.33455   0.560  0.575395
## OCCUR_HOUR13        -0.06109    0.30607  -0.200  0.841797
## OCCUR_HOUR14        -0.28319    0.32645  -0.867  0.385688
## OCCUR_HOUR15        -0.09660    0.29949  -0.323  0.747034
## OCCUR_HOUR16         0.06090    0.28468   0.214  0.830601
## OCCUR_HOUR17        -0.58662    0.34112  -1.720  0.085488 .

```

```
## OCCUR_HOUR18      -0.30940    0.27134   -1.140  0.254170
## OCCUR_HOUR19       0.15450    0.25255    0.612  0.540700
## OCCUR_HOUR20      -0.27115    0.27144   -0.999  0.317833
## OCCUR_HOUR21       0.05116    0.24679    0.207  0.835780
## OCCUR_HOUR22       0.05419    0.24194    0.224  0.822783
## OCCUR_HOUR23      -0.33466    0.27086   -1.236  0.216631
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 3002.9  on 4139  degrees of freedom
## Residual deviance: 2924.8  on 4104  degrees of freedom
## AIC: 2996.8
##
## Number of Fisher Scoring iterations: 13
```

```
# Predict probabilities (values between 0 and 1) with type set to response
model_data$PRED_PROB <- predict(nypd_model, type = "response")

# Create the comparison data for visualization of the model
comparison_data <- model_data |>
  # Group by location, then borough
  group_by(LOCATION, BORO) |>
  summarize(
    incident_count = n(),
    actual_pct_female = mean(VIC_SEX_NUM), # Average of 1s and 0s is the true proportion
    predicted_pct_female = mean(PRED_PROB), # Average model prediction for this group
    .groups = 'drop'
  ) |>
  # To eliminate the small sample variance noise, filter out the records
  # with < 5 incident counts
  filter(incident_count >= 5) |>
  mutate(
    # Calculate the raw residual
    residual = actual_pct_female - predicted_pct_female,

    # Calculate absolute error for easy sorting of biggest misses
    abs_err = abs(residual)
  ) |>
  # Sort by the largest absolute mistake first
  arrange(desc(abs_err))

# Lets see the top 10 biggest misses and catches
head(comparison_data, 10)
```

```
## # A tibble: 10 x 7
##   LOCATION BORO incident_count actual_pct_female predicted_pct_female residual
##   <fct>    <fct>         <int>         <dbl>         <dbl>         <dbl>
## 1 OTHER    MANH~             12          0.25          0.104         0.146
## 2 OTHER    BRONX             18           0          0.0899        -0.0899
## 3 TRANSIT  BRONX             29          0.276         0.190         0.0857
## 4 PLAYGRO~ BROO~             20          0.05          0.134        -0.0843
## 5 DWELLING STAT~             17          0.353         0.280         0.0731
```

```
## 6 DWELLING BRONX          96          0.115          0.186 -0.0715
## 7 VEHICLE QUEE~           9          0.111          0.171 -0.0594
## 8 PLAYGRO~ BRONX         11          0.182          0.123  0.0585
## 9 PLAYGRO~ MANH~         26          0.192          0.135  0.0574
## 10 TRANSIT BROO~         15          0.2          0.256 -0.0564
## # i 1 more variable: abs_err <dbl>
```

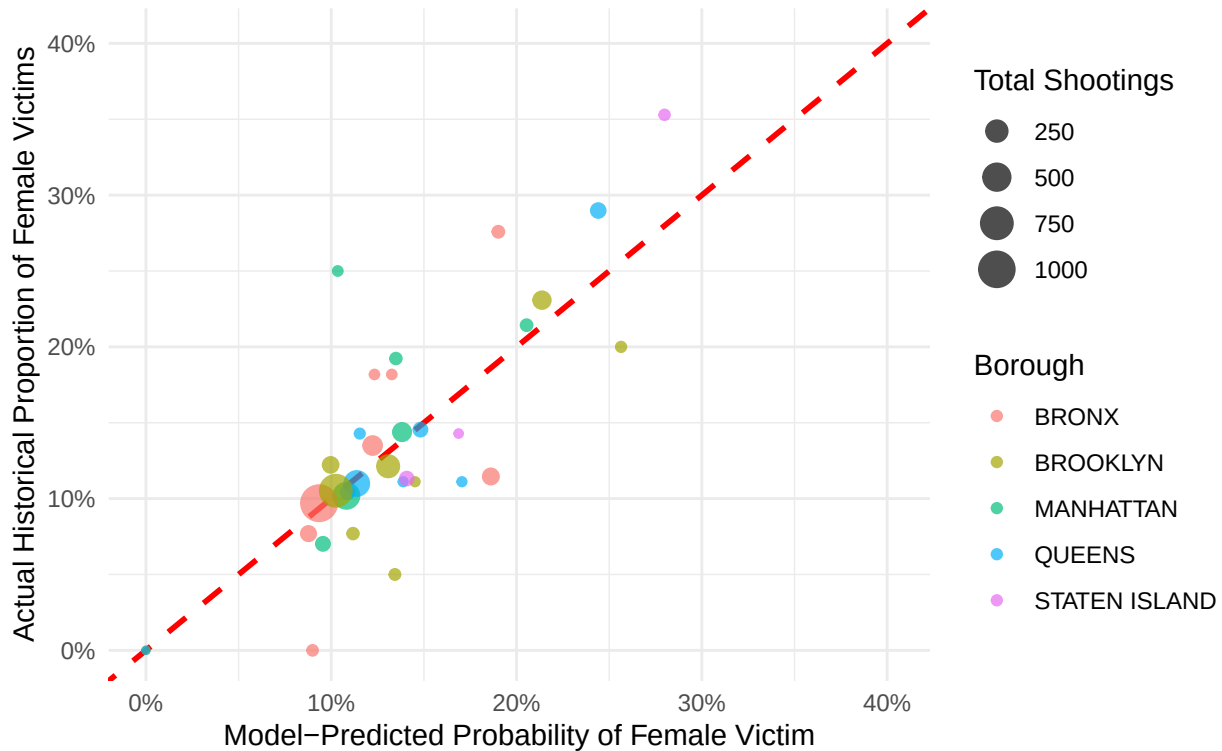
```
tail(comparison_data, 10)
```

```
## # A tibble: 10 x 7
##   LOCATION BORO incident_count actual_pct_female predicted_pct_female residual
##   <fct>    <fct>          <int>          <dbl>          <dbl>          <dbl>
## 1 DWELLING MANH~          28          0.214          0.205          8.87e-3
## 2 STREET   MANH~         432          0.102          0.108         -6.31e-3
## 3 HOUSING  MANH~         146          0.144          0.138          5.62e-3
## 4 COMMERC~ QUEE~          48          0.104          0.109         -4.75e-3
## 5 STREET   QUEE~         382          0.110          0.114         -3.83e-3
## 6 STREET   BRONX        1021          0.0970          0.0934          3.53e-3
## 7 STREET   BROO~         751          0.105          0.102          2.73e-3
## 8 HOUSING  QUEE~          55          0.145          0.148         -2.68e-3
## 9 PARKING~ BROO~           5           0          0.000000662 -6.62e-7
## 10 PARKING~ QUEE~           5           0          0.000000497 -4.97e-7
## # i 1 more variable: abs_err <dbl>
```

```
# Lets visualize the model evaluation with a scatter plot
ggplot(comparison_data, aes(x = predicted_pct_female, y = actual_pct_female)) +
# Prediction Reference line: Points on this line mean model prediction matches
# the actual perfectly
geom_abline(intercept = 0, slope = 1, color = "red", linetype = "dashed", linewidth = 1) +
# Plot the data points, sizing them by total incident count to show volume
geom_point(aes(size = incident_count, color = BORO), alpha = 0.7) +
# Format the scales
scale_x_continuous(
  labels = scales::percent, limits = c(0, max(comparison_data$actual_pct_female) + 0.05)) +
scale_y_continuous(
  labels = scales::percent, limits = c(0, max(comparison_data$actual_pct_female) + 0.05)) +
labs(
  title = "Model Evaluation: Predicted vs. Actual Female Victim Rates",
  subtitle = "Aggregated by Location and Borough (Groups with >= 5 incidents)",
  x = "Model-Predicted Probability of Female Victim",
  y = "Actual Historical Proportion of Female Victims",
  size = "Total Shootings",
  color = "Borough"
) +
theme_minimal() +
theme(
  plot.title = element_text(face = "bold", size = 14),
  legend.position = "right"
)
```

Model Evaluation: Predicted vs. Actual Female Victim Rates

Aggregated by Location and Borough (Groups with ≥ 5 incidents)



From the above scatter plot, we can see larger dots sitting directly on the prediction line around 10%. This means that for highest-volume space-and-location combinations (the big dots), the historical reality matches the model's predictions.

Conclusion

Overall, gun violence in New York City is heavily male-dominated across most hours and public locations. However, a closer look reveals two major exceptions: the proportion of shootings occurring inside homes (dwellings) and in transit systems is noticeably higher for female victims compared to male victims.

As a woman, choosing to analyze “Gun Violence Against Women” inherently brings a personal investment to the topic. To mitigate potential bias and ensure statistical objectivity, this study evaluated both male and female victims side-by-side across all times and locations. Furthermore, records with unknown entries (“U”) for victim sex were excluded to make sure missing data did not skew or distort the final results.

A major constraint of this study is that approximately 80% of the records lacked data for the specific location classification field (LOC_CLASSFCTN_DESC), primarily because the NYPD only began systematically recording this granular metric in recent years. While filtering out these records introduces an unavoidable selection bias, a demographic validation confirms that this selection bias did not skew the victim profile. Female victims account for 9.72% of the total historical data and remain remarkably consistent at 11.79% within the location-classified subset. This minimal variance demonstrates that the filtered subset remains highly representative of the broader dataset's victim demographics, rendering the observed trends a reliable indicator of how location dynamics differ between male and female victims.

AI Usage Statement

The Gemini 3.5 model developed by Google was utilized to brainstorm the visualization and statistical modeling options, ultimately guiding the selection of the included Stacked Bar Chart and Binomial GLM.

Session Info

```
## R version 4.6.1 (2026-06-24)
## Platform: aarch64-apple-darwin23
## Running under: macOS Sonoma 14.6
##
## Matrix products: default
## BLAS:   /Library/Frameworks/R.framework/Versions/4.6/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.6/Resources/lib/libRlapack.dylib; LAPACK version
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: America/Los_Angeles
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods   base
##
## other attached packages:
## [1] lubridate_1.9.5 forcats_1.0.1  stringr_1.6.0  dplyr_1.2.1
## [5] purrr_1.2.2     readr_2.2.0    tidyr_1.3.2    tibble_3.3.1
## [9] ggplot2_4.0.3   tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
## [1] gtable_0.3.6      compiler_4.6.1    tidyselect_1.2.1  scales_1.4.0
## [5] yaml_2.3.12       fastmap_1.2.0     R6_2.6.1          labeling_0.4.3
## [9] generics_0.1.4    knitr_1.51        pillar_1.11.1     RColorBrewer_1.1-3
## [13] tzdb_0.5.0        rlang_1.3.0       utf8_1.2.6        stringi_1.8.9
## [17] xfun_0.60         S7_0.2.2          ote1_0.2.0        timechange_0.4.0
## [21] cli_3.6.6         withr_3.0.3       magrittr_2.0.5    digest_0.6.39
## [25] grid_4.6.1        rstudioapi_0.19.0 hms_1.1.4         lifecycle_1.0.5
## [29] vctrs_0.7.3       evaluate_1.0.5    glue_1.8.1        farver_2.1.2
## [33] rmarkdown_2.31    tools_4.6.1       pkgconfig_2.0.3   htmltools_0.5.9
```